

18.06 CHEAT SHEET

$$\text{Rank} = \text{pivots} = \dim(C(A))$$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad-bc \neq 0$ then $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

LU Decomposition:

- Solving $A\vec{x} = \vec{b}$
1. Set $A=LU$ & substitute $Ax=B$
 2. Solve $LUx=B$ for x to solve the system.
 3. Let $Ux=C$.
 4. $LC=B$, $Ux=C$ so solve for $LC=B$ first using $c = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ and then solve $Ux=C$ for x .

L: Lower triangular
U: Upper triangular

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Computing Inverse By Elimination:

$$[A | I_p] \in \mathbb{R}^{p \times 2p} \xrightarrow{\text{Goal}} [I_p | A^{-1}]$$

Whatever you do on one side, do the same to the other side.

Matrix Operations:

- $A+B=B+A$
- $c(A+B)=cA+cB$
- $A+(B+C)=(A+B)+C$
- $D(A+B)=DA+DB$
- $(A+B)D=AD+BD$
- $A(BC)=(AB)C$
- $AB \neq BA$
- $AA^{-1}=I=A^{-1}A$
- $(A^T)^{-1}=(A^{-1})^T$

Uniqueness of Solution: $N(A)=\{\vec{0}\}$

- Uniqueness IFF the only solution to $A\vec{y}=\vec{0}$ is $\vec{y}=\vec{0}$ or $NS(A)=\{\vec{0}\}$.
- Uniqueness $\Leftrightarrow$ columns of A form a linearly independent family.

Kernel/Nullspace: To solve $A\vec{x}=\vec{0}$ Existence of Solution: $C(A)$

- 1) Eliminate A (Upper triangular)
- 2) Assign variables to free/nonpivot columns.
- 3) Plug into system of equations ($\vec{x}$ in terms of variables)

Image/Column Space: span of the columns of A

Elimination & picking the pivot columns

Ex: $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ span $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ $N(A)=N(L^{-1}A)$

Vector Subspace: (if subspace V of space W)

- 1) $\vec{0} \in V$
- 2) $\vec{u}, \vec{v} \in V, \vec{u}+\vec{v} \in V$
- 3) $\vec{u} \in V, c \in \mathbb{R}, c\vec{u} \in V$

$W + [\vec{0}]$ always a vector subspace of W .

Orthogonal Subspaces:

$$N(A)=C(A^T)^\perp \text{ and } C(A)=N(A^T)^\perp$$

Least Squares Approximation:

Line: $y=Cx+D$

$$(x_1, y_1) (x_2, y_2) (x_3, y_3) (x_4, y_4)$$

$$A = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \\ x_4 & 1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} C \\ D \end{bmatrix} \quad \vec{b} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

Coefficients of C, D

Resulting y values

$$\star A^T A \vec{x}^\star = A^T \vec{b} \star$$

If $A^T A$ is invertible: $\vec{x}^\star = (A^T A)^{-1} A^T \vec{b}$

Gram-Schmidt Process: (Converting to orthogonal bases)

① To orthogonal basis $(\vec{b}_1 \dots \vec{b}_q)$:

$$\begin{aligned} \vec{b}_1 &= \vec{a}_1 \\ \vec{b}_2 &= \vec{a}_2 - \frac{\vec{a}_2 \cdot \vec{b}_1}{\|\vec{b}_1\|^2} \vec{b}_1 \\ \vec{b}_3 &= \vec{a}_3 - \frac{\vec{a}_3 \cdot \vec{b}_1}{\|\vec{b}_1\|^2} \vec{b}_1 - \frac{\vec{a}_3 \cdot \vec{b}_2}{\|\vec{b}_2\|^2} \vec{b}_2 \end{aligned}$$

② To orthonormal basis $(\vec{c}_1 \dots \vec{c}_q)$:

$$\vec{c}_i = \frac{\vec{b}_i}{\|\vec{b}_i\|} \quad (\text{to force } \|\vec{c}_i\|=1)$$

$$A^2=0 \text{ if } C(A) \text{ is a subspace of } N(A). \quad A(Ax)=0=A^2x$$

Linear Maps:

A linear map $f: V \rightarrow W$ satisfies:

$$f(\vec{x}+\vec{y}) = f(\vec{x}) + f(\vec{y})$$

$$f(c\vec{x}) = c f(\vec{x})$$

Adjoint linear map $\rightarrow f: \mathbb{R}^q \mapsto \mathbb{R}^p$ (linear map)

the adjoint of f is $f^*: \mathbb{R}^p \mapsto \mathbb{R}^q$

$$\forall \vec{x} \in \mathbb{R}^q, \vec{y} \in \mathbb{R}^p: f(\vec{x}) \cdot \vec{y} = \vec{x} \cdot f^*(\vec{y})$$

(dot product) inner product $\mathbb{R}^q \mathbb{R}^p$ (Transpose)

Ex: $f: \mathbb{R}^2 \mapsto \mathbb{R}^3$

$$(x, y) \mapsto (2x+y, 0, 3y)$$

1) Canonical basis of $\mathbb{R}^2$ to canonical basis of $\mathbb{R}^3$

2) Compute $f(1,0) = (2,0,0)$

$$f(0,1) = (1,0,3)$$

3) Write result as linear combination of canonical basis of $\mathbb{R}^3$

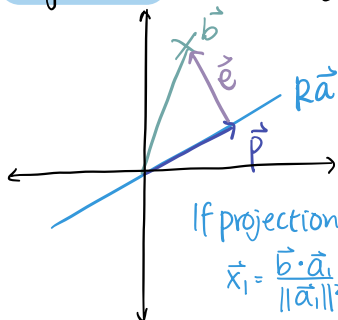
$$f(1,0) = a(1,0,0) + b(0,1,0) + c(0,0,1) = (2,0,0)$$

$$f(0,1) = e(1,0,0) + f(0,1,0) + g(0,0,1) = (1,0,3)$$

$$\begin{bmatrix} a & e \\ b & f \\ c & g \end{bmatrix} \quad \begin{matrix} a=2 & b=0 & c=0 \\ e=1 & f=0 & g=3 \end{matrix}$$

(linear map in basis of $\mathbb{R}^3$)

Projection:



$$\vec{e} = \vec{b} - \vec{p}$$

$$\vec{p} = A\vec{x}^\star$$

$$\vec{p} = A(A^T A)^{-1} A^T \vec{b}$$

$$\text{Projection of } \vec{b} \text{ onto } \vec{a}: \vec{p} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2} \vec{a}$$

If projection of $\vec{b}$ onto $\text{span}(\vec{a}_1, \vec{a}_2)$: $\vec{p} = x_1 \vec{a}_1 + x_2 \vec{a}_2$

$$x_1 = \frac{\vec{b} \cdot \vec{a}_1}{\|\vec{a}_1\|^2} \quad x_2 = \frac{\vec{b} \cdot \vec{a}_2}{\|\vec{a}_2\|^2}$$

*This only works if we have an orthonormal basis.

Orthogonal Projections $\rightarrow \vec{p} = Q Q^T \vec{b}$ columns are vectors of the orthonormal basis

$$A = [\vec{a}_1 \dots \vec{a}_q]$$

Determinant Properties:

* Changes to rows + columns are both allowed.

Inverse of a Matrix:

$$A^{-1} = \frac{1}{\det A} C^T$$

A Stochastic matrix $A \in \mathbb{R}^{p \times p}$ satisfies:

1) It has positive values.

2) The sum of the column coefficients is 1.

1) $\det(I_p) = 1$

2) $\det(A') = -\det(A)$ if A' is obtained from exchanging 2 rows of A .

3)
$$\det \begin{bmatrix} \lambda R_1 + \mu R_1' \\ R_2 \\ \vdots \\ R_p \end{bmatrix} = \lambda \det \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_p \end{bmatrix} + \mu \det \begin{bmatrix} R_1' \\ R_2 \\ \vdots \\ R_p \end{bmatrix}$$

4) If 2 rows are equal, $\det(A) = 0$.

5) Subtracting one row to the other doesn't change $\det(A)$

6) If a full row is 0, then $\det(A) = 0$.

7) If A is a triangular matrix,
 $\det(A) = \text{product of the diagonal terms.}$

8) $\det(A) \neq 0$ IFF A is invertible.

9) $\det(AB) = \det(A) \cdot \det(B)$

10) $\det(A^T) = \det(A)$

Diagonalization:

$A \vec{b}_i = \lambda \vec{b}_i$ $B = [\vec{b}_1 \dots \vec{b}_p]$ $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix}$ $D^{-1} = \begin{bmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_p^{-1} \end{bmatrix}$

$\{\vec{b}_1, \dots, \vec{b}_p\}$ forms some basis s.t. D is diagonalized matrix of A .

$A = BDB^{-1}$

• If A has p distinct eigenvalues, then A is diagonalizable.

• $A\vec{x} = \lambda\vec{x} \rightarrow A^2\vec{x} = \lambda^2\vec{x}, \dots, A^k\vec{x} = \lambda^k\vec{x}$ ($k > 0$)

If invertible, then also true for $k < 0$

$A = BDB^{-1} \leadsto A^k = \underbrace{BDB^{-1} \dots BDB^{-1}}_{k \text{ times}} = BD^k B^{-1}$

$D^k = \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_p^k \end{bmatrix}$

ODEs: $\{\vec{v}_1, \dots, \vec{v}_p\}$ eigenvectors

$\frac{d}{dt} \vec{x}(t) = A\vec{x}(t)$ $\vec{x}(t) = c_1 e^{t\lambda_1} \vec{v}_1 + \dots + c_p e^{t\lambda_p} \vec{v}_p$

$\vec{x}(0) = \vec{x}_0$ $\vec{x}(0) = c_1 \vec{v}_1 + \dots + c_p \vec{v}_p = \vec{x}_0$

Singular Value Decomposition (SVD):

$A: m \times n$ matrix.

$\rightarrow \exists$ two orthogonal matrices $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$ such that

$Q^{-1}AP = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \end{pmatrix}$

$A = \underbrace{U}_{m \times m} \underbrace{\Sigma}_{m \times n} \underbrace{V^T}_{n \times n}$ (columns of V , not V^T)

A matrix $A \in \mathbb{R}^{p \times p}$ is symmetric positive definite if one of the property satisfied:

1) eigenvalues are positive (nonnegative)

2) $\forall \vec{x} \neq \vec{0}, \vec{x}^T A \vec{x} > 0$ ≥ 0

3) $\exists$ invertible matrix $R \in \mathbb{R}^{p \times p}$ s.t. $A = R^T R$

doesn't have to be invertible for positive semi definite.

* A has to be symmetric for us to consider positive definite.

For nonzero LSVs, you can take $A(RSV) = LSV$ (image of A using right singular vector) $\rightarrow$ then, normalize

(How to find P, Q, σ_i ?) To check solution: $A\vec{v}_i = \sigma_i \vec{w}_i$

1. $A^T A$ always a square matrix
 $\rightarrow$ find eigenvalue $\lambda_1, \lambda_2, \dots, \lambda_r, 0, \dots, 0$
positive

* trick: we can also find the eigenvalue of AA^T
 $\rightarrow \sigma_i = \sqrt{\lambda_i}$

2. Find the orthonormal eigenvectors of $A^T A$.
 $\rightarrow \vec{v}_1, \dots, \vec{v}_n$
 $\rightarrow P = (\vec{v}_1 \dots \vec{v}_n)$ Right singular vectors

* Normalize vectors

3. Look at $A\vec{v}_i$. Normalize $A\vec{v}_1, \dots, A\vec{v}_r \rightarrow \vec{w}_1, \dots, \vec{w}_r = \ker(A^T)$ left singular vectors
Find an orthonormal basis of $\ker(AA^T) = \vec{w}_{r+1}, \dots, \vec{w}_m$.
orthonormal eigenvectors of AA^T